BLOWING UP IN THE PRODUCT OF THE CENTER WITH A CARTIER DIVISOR CHANGES NOTHING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The first of the two assertions is proved by name-combining two lemmas, the second by an in-place algebra argument; the card is taken and the shape of part (1) is recorded here for the reader.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Blowing up two ideals and the effective Cartier divisor case. Lemmas lemma-blowing-up-two-ideals (8588-8595) and lemma-blow-up-effective-Cartier-divisor (8505-8509), used at proof lines 8952-8954. Give the first assertion.
- The affine description of a blowup. Lemma lemma-blowing-up-affine, used at proof lines 8954-8956. Translates the second assertion into the algebra problem.

Source reference. Original lemma 0807: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10172–10172. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0804: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10169–10169. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 080A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10175–10175. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $Z \subset S$ be a closed subscheme. Let $D \subset S$ be an effective Cartier divisor. Let $Z' \subset S$ be the closed subscheme cut out by the product of the ideal sheaves of Z and D. Let $S' \to S$ be the blowup of S in Z.*

(1) *The blowup of S in Z' is isomorphic to $S' \to S$.*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *Let $f : X \to S$ be a morphism of schemes and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. If $\mathcal{F}$ has no nonzero local sections supported in $f^{-1}D$, then the strict transform of $\mathcal{F}$ relative to the blowing up in Z agrees with the strict transform of $\mathcal{F}$ relative to the blowing up of S in Z'.*

Proof. The first statement follows on combining Lemmas 080A and 0807. Using Lemma 0804 the second statement translates into the following algebra problem. Let A be a ring, $I \subset A$ an ideal, $x \in A$ a nonzerodivisor, and $a \in I$. Let M be an A-module whose x-torsion is zero. To show: the a-power torsion in $M \otimes_A A[\frac{I}{a}]$ is equal to the xa-power torsion. The reason for this is that the kernel and cokernel of the map $A \to A[\frac{I}{a}]$ is a-power torsion, so this map becomes an isomorphism after inverting a. Hence the kernel and cokernel of $M \to M \otimes_A A[\frac{I}{a}]$ are a-power torsion too. This implies the result. □